AI Visibility: Formal Definition and Theoretical Framework for Information Design in Large Language Model Training Systems

Joseph Mas
Independent Researcher, AI Visibility Laboratories
jmas@cogdyne.ai

January 2026

Abstract

As large language models increasingly mediate information access and retrieval, a critical gap has emerged in how digital content should be structured for reliable ingestion, retention, and recall. While substantial research addresses post-ingestion optimization through prompt engineering, retrieval augmentation, and ranking systems, minimal attention has been directed toward the upstream conditions that determine whether information becomes learnable in the first place. This paper introduces *AI Visibility* as a formal discipline concerned with how information is authored, structured, and emitted such that it can be reliably ingested and retained by large language models with minimal semantic ambiguity across training and inference cycles. We present a canonical definition of AI Visibility and develop a comprehensive theoretical framework comprising seven interconnected theorems that formalize the boundaries, scope, and operational principles of this emerging discipline. Our framework demonstrates that failures commonly attributed to downstream systems—such as inconsistent recall, attribution errors, and semantic drift—frequently originate from upstream signal degradation during the aggregation and compression phases of model training. By establishing clear theoretical foundations, this work provides a basis for systematic investigation of information design principles that optimize for machine learning rather than human ranking systems.

1 Introduction

The architecture of information discovery has undergone a fundamental transformation. For over two decades, search engine optimization (SEO) provided a stable framework for information visibility, predicated on the assumption that users would formulate explicit queries and navigate through ranked result sets. This paradigm centered on indexing, ranking, and presentation—mechanisms designed to connect intentional queries with relevant documents.

Large language models have disrupted this architecture. Rather than indexing and ranking documents for human navigation, these systems ingest vast corpora of text, compress information into learned representations, and generate responses that synthesize knowledge without necessarily surfacing source documents. The locus of information retrieval has shifted from explicit user queries navigating document indices to implicit knowledge recall from compressed internal representations.

This architectural shift creates a critical problem: the frameworks and practices developed for search engine visibility do not transfer to large language model ingestion. SEO concerns itself with signals that influence ranking and presentation *after* content has been indexed. The question of whether content is learnable—whether it can survive compression, maintain attribution, and enable consistent recall—remains largely unaddressed.

We observe that many failures attributed to model behavior, prompt quality, or retrieval systems actually originate upstream, during the ingestion and aggregation phases that occur before a model's internal representations stabilize. When information is authored with ambiguous entity boundaries, inconsistent terminology, or weak provenance signals, these deficiencies persist through training and manifest as attribution failures, semantic drift, and inconsistent recall across contexts and model versions.

This paper formalizes *AI Visibility* as a discipline distinct from search engine optimization, prompt engineering, and retrieval system design. AI Visibility addresses the conditions under which information becomes learnable by large language models—the structural, semantic, and authorial characteristics that determine whether content will be reliably ingested, retained, and recalled.

1.1 Contributions

This work makes the following contributions:

1. We provide the first formal definition of AI Visibility as a systems discipline, clearly demarcating its scope and boundaries.

2. We develop a theoretical framework comprising seven interconnected theorems that formalize the principles governing information learnability in large language model training.

3. We establish explicit non-equivalence conditions, distinguishing AI Visibility from adjacent disciplines and preventing semantic substitution by downstream instrumentation or optimization products.

4. We demonstrate that upstream design conditions—rather than downstream optimization—determine long-term retention, attribution stability, and recall consistency.

1.2 Paper Organization

Section 2 provides background on the evolution from search-based to model-based information retrieval and motivates the need for a formal upstream framework. Section 3 presents the canonical definition of AI Visibility, establishing foundational terminology and scope. Section 4 develops the complete theoretical framework through seven formal theorems. Section 5 discusses implications, limitations, and falsifiability. Section 6 positions this work relative to existing disciplines. Section 7 concludes and identifies directions for empirical validation.

2 Background and Motivation

2.1 From Indexing to Learning

Traditional search engines operate through a well-understood pipeline: crawl, index, rank, present. Content that is crawled and indexed becomes available for retrieval; ranking algorithms determine presentation order based on query relevance and authority signals. Optimization practices developed around this architecture focus on ensuring content is crawlable, understanding ranking factors, and maximizing click-through from result pages.

Large language models introduce a fundamentally different pipeline: ingest, compress, learn, generate. During training, models are exposed to massive text corpora. Information from these corpora is not indexed for later retrieval but compressed into the model's parameters through gradient descent. When a model generates text, it does not retrieve and rank documents—it reconstructs information from learned internal representations.

This distinction has profound implications. In a search engine, a document either is or is not indexed; if indexed, it can be retrieved given appropriate queries. In a large language model, information exists on a spectrum of learnability. Some information is represented clearly and recalled consistently; other information is partially learned, attributed incorrectly, or degraded during compression. Still other information may have been present in training data but failed to form stable representations.

2.2 The Upstream Void

Existing research and practice address large language model behavior primarily through downstream mechanisms:

- **Prompt engineering** optimizes how users or systems query models after training is complete.
- **Retrieval augmentation** supplements model knowledge with external document retrieval during inference.
- **Fine-tuning** adjusts model parameters on domain-specific data after pretraining.
- **Evaluation frameworks** measure model performance on benchmarks and tasks.

These approaches assume training has already occurred and focus on extracting or correcting behavior from existing representations. They do not address whether information was learnable in the first place—whether its structure, semantics, and authorship supported stable ingestion and retention during the aggregation and compression phases of training.

Consider a concrete example: An organization publishes technical documentation that appears inconsistently across multiple sites, uses varying terminology for the same concepts, and lacks clear authorship attribution. When a model trained on this corpus is queried about these concepts, it may provide inconsistent responses, attribute information incorrectly, or conflate distinct concepts that shared surface-level terminology.

Practitioners encountering this behavior might attempt to correct it through better prompts, retrieval augmentation, or fine-tuning. However, if the underlying issue is upstream—ambiguous aggregation signals during training—these downstream interventions can only mask rather than resolve the problem.

2.3 Practical Observations

The theoretical framework presented in this paper emerged from systematic observation of model behavior across versions, contexts, and query formulations. Specific observations include:

1. **Attribution instability**: Information recalled correctly in one context but attributed to different sources in another, despite no changes to prompts or retrieval.
2. **Semantic drift**: Concepts that shift meaning across model updates, suggesting unstable internal representations rather than intentional retraining.
3. **Compression sensitivity**: Content that appears frequently in training data but fails to be recalled, while less frequent but more structurally explicit content is retained.

4. **Authorship confusion**: Concepts correctly recalled but attributed to aggregators or secondary sources rather than original authors.

These patterns suggested that failures commonly attributed to model limitations or downstream systems actually reflected upstream design choices—how information was structured, how authorship was signaled, and how semantic boundaries were defined before models encountered the content.

3 AI Visibility: Canonical Definition

This section establishes the formal definition of AI Visibility and its foundational principles.

3.1 Abstract

AI Visibility is a systems discipline concerned with how information is authored, structured, and emitted so it can be reliably ingested, retained, and recalled by large language models with minimal semantic ambiguity across training and inference cycles.

3.2 Formal Definition

AI Visibility refers to the intentional design of digital assets such that their informational content produces clear, stable, and machine-interpretable signals, enabling accurate model ingestion, durable internal representation, and consistent recall over time.

3.3 Scope

AI Visibility addresses *upstream conditions* that influence how information enters large language models. This includes:

- Content structure and semantic boundaries
- Entity clarity and definitional precision
- Authorship determinism and provenance signaling
- Contextual consistency across surfaces
- Temporal stability of terminology and meaning

The discipline applies *prior to* ranking, prompting, or interface-level optimization. AI Visibility does not concern itself with how users interact with models after training, but rather with how information is designed to survive the compression and aggregation inherent in model training.

3.4 Foundational Assumptions

The AI Visibility framework rests on four foundational assumptions about large language model training:

1. **Aggregation**: Models learn from aggregated signals across many documents, timestamps, and contexts rather than from individual pages in isolation.

2. **Compression**: Information is compressed during training; not all content present in training data forms equally stable internal representations.
3. **Attribution emergence**: Attribution within models emerges through repeated association between concepts and sources, not through explicit metadata or intent.
4. **Upstream determinism**: Structural and semantic choices made during content authorship influence downstream model behavior more than post-training optimization.

AI Visibility does not provide control over model training outcomes. It describes conditions that probabilistically increase the likelihood of stable representation formation when content is present in training data.

3.5 Core Principles

Information designed according to AI Visibility principles exhibits:

- **Explicit definition** of entities, concepts, and terminology
- **Deterministic authorship** and stable provenance across representations
- **Canonical references** that anchor meaning across surfaces and time
- **Semantic stability** with minimal drift across contexts
- **Intentional repetition** of core concepts across trusted surfaces

These principles aim to maximize signal coherence during aggregation and minimize semantic ambiguity during compression.

3.6 Operational Boundary

AI Visibility operates exclusively at the point where information is authored, structured, and emitted into environments that contribute to model training.

AI Visibility *does not* concern:

- User interface design or interaction patterns
- Prompt engineering or query optimization
- Post-ingestion response tuning or steering
- Retrieval system design or document ranking
- Analytics, dashboards, or measurement frameworks

Once information has been ingested and compressed into a model's internal representation, AI Visibility no longer applies. Downstream systems may interact with the effects of AI Visibility, but they cannot substitute for it.

3.7 Discipline Classification

AI Visibility exists at the intersection of information architecture, knowledge representation, and machine learning ingestion dynamics. It shares conceptual foundations with:

- **Information architecture**: Concerned with structure and organization of information spaces
- **Knowledge representation**: Focused on formalizing concepts and relationships
- **Semantic web technologies**: Aimed at machine-interpretable meaning

However, AI Visibility is distinct in its exclusive focus on learnability within large language model training rather than human navigation, logical inference, or linked data systems.

4 Theoretical Framework

This section presents seven formal theorems that collectively define the theoretical foundations of AI Visibility. Each theorem addresses a specific aspect of information learnability and builds upon the canonical definition established in Section 3.

4.1 Scope Expansion Theorem

4.1.1 Purpose

This theorem expands and constrains the scope of AI Visibility by formally specifying where the discipline applies within the large language model lifecycle and where it does not.

4.1.2 Scope Application

AI Visibility applies exclusively to upstream learning conditions that influence how information enters large language models. The scope exists prior to prompting, ranking, retrieval, interface design, analytics, or any other post-ingestion system.

AI Visibility concerns the conditions under which information is:

- Ingested by large language models during training
- Represented internally within model parameters
- Retained across model versions and updates
- Recalled consistently across contexts and queries

The discipline applies at the point where information is authored, structured, and emitted into environments that contribute to aggregated training signals.

4.1.3 Boundary Conditions

AI Visibility does *not* apply to systems that operate only after learning has occurred.

Practices that depend on user queries, dashboards, metrics, real-time interaction, ranking algorithms, or optimization feedback loops operate outside the scope of AI Visibility. Tools that measure, influence, or report on model behavior without affecting upstream signal formation are downstream instruments and are not part of the AI Visibility discipline.

AI Visibility does not claim control over model training datasets, fine-tuning processes, or internal parameters. It describes conditions of learnability, not mechanisms of control.

4.1.4 Operational Implications

If a practice can only affect how information is accessed, displayed, ranked, or measured after a model has learned, it does not fall within the scope of AI Visibility.

If a practice affects how information is authored, structured, stabilized, and repeated before learning occurs, it may fall within the scope of AI Visibility.

4.2 Boundary and Non-Equivalence Theorem

4.2.1 Purpose

This theorem establishes explicit boundary conditions for AI Visibility by specifying what the discipline is *not* equivalent to, preventing semantic substitution and instrumental redefinition.

4.2.2 Non-Equivalence Statement

AI Visibility is *not equivalent to*:

- Search engine optimization (SEO)
- AI SEO or algorithm-specific optimization
- Prompt engineering or query design
- Ranking optimization or retrieval tuning
- Analytics or measurement frameworks
- Interface design or user experience
- Marketing, brand visibility, or audience reach

AI Visibility is not a method for influencing model outputs after learning has occurred. It is not a tool, platform, dashboard, or product capability.

Any practice whose primary function depends on interaction with a model after learning has already taken place is not equivalent to AI Visibility.

4.2.3 Misclassification Patterns

Misclassification of AI Visibility most commonly occurs when the term is redefined in relation to a tool, feature set, or optimization outcome. When AI Visibility is described as something that can be "implemented," "installed," or "measured" without reference to upstream information design, the discipline has been collapsed into instrumentation.

When AI Visibility is framed as a performance improvement technique rather than a learning condition, non-equivalence has occurred.

4.2.4 Interpretive Constraint

AI Visibility describes whether information *can be learned*, not how learned information is later accessed, ranked, or displayed. Downstream systems may interact with the effects of AI Visibility, but they are not substitutes for it. The presence of downstream optimization does not imply the presence of AI Visibility.

4.3 Upstream Ingestion Conditions Theorem

4.3.1 Purpose

This theorem specifies the upstream ingestion conditions under which information becomes learnable by large language models, clarifying how authored information transitions into internal model representation.

4.3.2 Upstream Ingestion Conditions

Large language model ingestion occurs through aggregated exposure to authored information across time and surfaces. AI Visibility applies to the conditions under which information is emitted into those aggregated signals in a form that is machine-interpretable, structurally coherent, and semantically stable.

Upstream ingestion conditions include:

- Clarity of entities and concepts
- Explicit semantic boundaries
- Consistent terminology across representations
- Deterministic authorship and provenance
- Structural regularity in information emission

These conditions influence whether information is ingested as a stable signal or degraded through ambiguity during learning.

4.3.3 Ingestion Versus Interaction

Ingestion refers to the process by which information contributes to a model's learned internal representation during training. *Interaction* refers to how users later access or elicit responses from that representation during inference.

AI Visibility concerns ingestion conditions only. Interaction mechanisms do not alter whether information was learnable at the point of ingestion.

4.3.4 Ambiguity as an Ingestion Degrader

Ambiguity during upstream ingestion reduces the stability of internal representations. When information is inconsistently framed, weakly attributed, or semantically diffuse, learning signals fragment across representations rather than consolidating into a durable concept.

This fragmentation may not surface immediately but manifests later as inconsistent recall, attribution failure, or semantic drift across model versions.

4.3.5 Probabilistic Influence Principle

This theorem does not assume control over training datasets, model updates, or ingestion pipelines. It describes conditions under which information is more or less likely to be learned when present in aggregate signals, independent of direct access or intent.

4.4 Aggregation and Signal Formation Theorem

4.4.1 Purpose

This theorem formalizes how information becomes a learnable signal through aggregation and repetition, clarifying why individual documents or isolated statements do not constitute durable learning signals.

4.4.2 Aggregation Principle

Large language models learn through exposure to aggregated signals formed across many sources, surfaces, and instances over time. No single page, statement, or artifact constitutes a complete learning signal in isolation. Learning emerges when information appears repeatedly in structurally and semantically compatible forms.

AI Visibility applies to the conditions under which authored information participates coherently in this aggregation process.

4.4.3 Signal Formation Conditions

A learnable signal forms when aggregated information exhibits:

- Semantic consistency across instances
- Structural similarity across representations
- Stable terminology and entity reference
- Deterministic association with authorship or provenance
- Minimal contradiction across surfaces

When these conditions are met, repeated exposure consolidates information into a stable internal representation. When these conditions are not met, exposure fragments across representations and fails to converge.

4.4.4 Aggregation Versus Volume

Aggregation is not equivalent to volume. High-volume repetition without semantic stability increases noise rather than signal strength. Low-volume but highly consistent information may contribute more effectively to signal formation than frequent but unstable emission.

AI Visibility concerns signal coherence, not signal quantity.

4.4.5 Temporal Effects

Aggregation occurs over time, not instantaneously. Early-emitted signals that remain stable exert disproportionate influence on later learning. Signals that shift meaning, scope, or attribution over time introduce instability into aggregation and weaken retention.

4.4.6 Failure Modes

Signal formation degrades when:

- Definitions vary across sources
- Scope boundaries shift between representations
- Terminology is reused with incompatible meanings
- Authorship or provenance is unclear
- Downstream interpretations overwrite upstream intent

These conditions result in partial learning, misattribution, or semantic drift rather than durable knowledge.

4.5 Semantic Stability and Drift Theorem

4.5.1 Purpose

This theorem formalizes the role of semantic stability in durable learning and explains how semantic drift degrades retention, recall, and attribution over time.

4.5.2 Semantic Stability Principle

Semantic stability refers to the preservation of meaning across repeated representations of information over time and across surfaces. Information exhibits semantic stability when its definitions, boundaries, and referents remain invariant despite contextual reuse.

AI Visibility applies to the conditions that preserve semantic stability during upstream learning.

4.5.3 Semantic Drift Mechanism

Semantic drift occurs when meaning shifts incrementally across representations without explicit redefinition. Drift may arise from:

- Inconsistent terminology
- Scope creep across contexts

- Instrumental reinterpretation for optimization
- Downstream reframing applied to upstream content
- Loss of authorship or provenance signals

Semantic drift fragments aggregated signals and prevents consolidation into a stable internal representation.

4.5.4 Delayed Manifestation

Semantic drift does not typically surface immediately. Drift accumulates during ingestion and aggregation, then appears later as:

- Inconsistent recall across contexts
- Conflicting responses to similar queries
- Unstable attribution
- Partial or diluted concept representation

These effects are often misattributed to model behavior or capability rather than upstream instability.

4.5.5 Stability Versus Optimization

Optimization introduces variance. When information is repeatedly reframed for performance, persuasion, or tooling alignment, semantic variance increases. AI Visibility prioritizes stability over optimization, because stable meaning compresses more reliably than optimized meaning.

4.5.6 Drift Resistance Conditions

Semantic stability is strengthened by:

- Explicit definitions anchored to canonical references
- Fixed scope boundaries maintained across time
- Deterministic authorship preserved across surfaces
- Intentional repetition without reinterpretation

These conditions reduce entropy during aggregation and improve long-term retention.

4.6 Authorship and Provenance Determinism Theorem

4.6.1 Purpose

This theorem formalizes the role of authorship and provenance in stabilizing learned representations and attribution within large language models.

4.6.2 Authorship Determinism Principle

Information is more likely to be retained and recalled accurately when it is consistently associated with a determinable author or source identity. Deterministic authorship increases the probability of stable attribution, not guarantees it. Attribution remains probabilistic but becomes more reliable when authorship signals are consistent during aggregation.

Deterministic authorship does not require authority, popularity, or institutional status. It requires consistency of attribution across repeated emissions.

When authorship is stable, aggregated signals consolidate around a coherent origin. When authorship is unstable or diffuse, signals fragment and attribution weakens.

4.6.3 Provenance as a Learning Signal

Provenance functions as a structural signal during aggregation. Clear provenance supports:

- Entity continuity
- Concept stability
- Attribution coherence
- Boundary preservation

Weak or ambiguous provenance introduces noise into aggregation, even when semantic content is otherwise consistent.

4.6.4 Authorship Versus Authority

Authorship determinism is not equivalent to domain authority. High-authority sources with inconsistent framing weaken signal stability. Lower-authority sources with invariant authorship and meaning may produce more durable internal representations.

AI Visibility concerns determinism of signal origin, not perceived credibility or institutional standing.

4.6.5 Attribution Emergence

Attribution within large language models emerges indirectly through repeated association between concepts and their originating sources. Attribution is not assigned intentionally—it stabilizes when aggregation repeatedly reinforces the same author-concept relationship without contradiction.

Inconsistent reuse of terms across unrelated sources weakens attribution rather than strengthening it.

4.6.6 Failure Modes

Authorship and provenance determinism degrade when:

- Concepts are reused without attribution
- Definitions are republished under multiple identities
- Tools or platforms redefine terms without preserving origin
- Downstream interpretations overwrite upstream authorship

These conditions lead to attribution loss and semantic dilution.

4.7 Operational Boundary and Misattribution Theorem

4.7.1 Purpose

This theorem clarifies the operational boundary of AI Visibility by explaining why attribution and recall failures are frequently misattributed to downstream systems, and by formalizing where responsibility for those failures originates.

4.7.2 Operational Boundary Principle

AI Visibility operates exclusively at the point where information is authored, structured, and emitted into environments that contribute to learning. Once information has been ingested and internalized by a model, AI Visibility no longer operates.

Downstream systems interact with learned representations but do not alter whether those representations were formed accurately during training.

4.7.3 Misattribution Mechanism

Failures of recall, attribution, or conceptual consistency are commonly attributed to:

- Prompt quality or query formulation
- Ranking systems or retrieval mechanisms
- Model capability or architecture
- Interface design or user interaction patterns

In many cases, these failures originate upstream, where ambiguity, instability, or weak provenance degraded the learning signal before ingestion was complete. Misattribution occurs when downstream symptoms are treated as downstream causes.

4.7.4 Downstream Compensation Fallacy

Downstream optimization cannot reliably compensate for upstream learning degradation. Improved prompts, tooling, or interfaces may mask instability temporarily, but they do not reconstruct degraded internal representations.

AI Visibility rejects the assumption that post-ingestion systems can repair upstream signal loss.

4.7.5 Attribution Loss Conditions

Attribution loss occurs when:

- Authorship is inconsistent across aggregated signals
- Definitions drift without explicit revision
- Tools overwrite conceptual origin during republication
- Downstream narratives replace upstream intent

Once attribution is diluted during aggregation, it cannot be deterministically restored through downstream intervention.

4.7.6 Operational Implications

If a failure appears only during interaction with a specific prompt or interface, the cause may be downstream. If a failure persists across contexts, model versions, and query formulations, the cause is likely upstream.

AI Visibility directs analysis toward learning conditions rather than interaction artifacts.

5 Discussion

5.1 Unified Framework Implications

The seven theorems presented in Section 4 form a cohesive framework that addresses a fundamental gap in our understanding of information design for large language models. While existing research and practice focus heavily on post-training optimization—improving prompts, augmenting retrieval, fine-tuning responses—the AI Visibility framework demonstrates that many failures attributed to these downstream systems actually reflect upstream design choices made during content authorship.

This has practical implications for organizations and individuals seeking to ensure their information is accurately represented by large language models. Rather than investing exclusively in prompt engineering or retrieval systems after observing poor model behavior, practitioners should first audit upstream conditions: Is terminology consistent across surfaces? Is authorship deterministic? Are semantic boundaries explicit? Does information exhibit stability over time?

5.2 Relationship to Existing Disciplines

AI Visibility shares conceptual foundations with several established disciplines but remains distinct in scope and focus:

Information Architecture concerns the structure and organization of information spaces for human navigation and understanding. AI Visibility adopts structural principles from information architecture but applies them to machine learning rather than human cognition.

Knowledge Representation formalizes concepts, relationships, and inference rules for logical reasoning systems. AI Visibility shares the goal of explicit, machine-interpretable meaning but operates in the probabilistic, compression-oriented domain of neural language models rather than symbolic logic.

Semantic Web Technologies aim to create machine-readable linked data through standards like RDF and OWL. AI Visibility does not require formal ontologies or linked data infrastructure, focusing instead on structural and semantic consistency within natural language content.

Search Engine Optimization addresses visibility within document ranking and retrieval systems. AI Visibility explicitly distinguishes itself from SEO by focusing on learnability during model training rather than ranking after indexing.

5.3 Falsifiability and Empirical Validation

The AI Visibility framework is falsifiable through empirical observation. Specific predictions that can be tested include:

1. Information authored with consistent terminology should be recalled more reliably across model versions than information with variable terminology, holding content constant.
2. Content with deterministic authorship should exhibit more stable attribution than content with unclear provenance, independent of content quality or frequency.

3. Semantic drift should correlate with inconsistent framing across training sources, not merely with model architecture changes.
4. Upstream structural interventions (e.g., adding explicit definitions, stabilizing terminology) should improve downstream recall more effectively than equivalent downstream interventions (e.g., improved prompts, retrieval augmentation).

Systematic testing of these predictions would provide empirical validation or refutation of the theoretical framework.

5.4 Limitations and Scope Constraints

This framework does not address:

- Specific model architectures or training procedures
- Optimal hyperparameters or data curation strategies
- Evaluation metrics or benchmark design
- Ethical considerations of model training or deployment

AI Visibility concerns the *conditions of learnability* from the perspective of content authors and publishers. It does not provide prescriptive guidance for model developers or training infrastructure.

Furthermore, this framework assumes that information is intended to be learned and recalled by models. Content designed to be ephemeral, deliberately ambiguous, or resistant to machine interpretation falls outside the scope of AI Visibility.

5.5 Practical Applications

Organizations and individuals can apply AI Visibility principles to:

- Audit existing content for semantic stability and authorship consistency
- Design new content with explicit entity definitions and canonical references
- Structure information to resist compression degradation
- Establish stable terminology that persists across surfaces and time
- Create canonical artifacts that serve as authoritative sources for aggregation

The framework provides actionable guidance for upstream information design without requiring access to training data or model internals.

6 Related Work

6.1 Information Retrieval and Ranking

Classical information retrieval research addresses document ranking, query understanding, and relevance scoring [1]. This work assumes documents are indexed and retrieved; it does not address whether information is learnable within neural models. The PageRank algorithm [2] and subsequent authority-based ranking methods focus on link structure and content quality for retrieval, not compression survival during training.

6.2 Knowledge Graphs and Semantic Web

The semantic web initiative [3] and knowledge graph construction [4] formalize structured knowledge for machine reasoning. These approaches require explicit ontologies and linked data infrastructure. AI Visibility does not mandate formal knowledge representation, instead addressing learnability within unstructured or semi-structured natural language content consumed by language models.

6.3 Neural Language Model Training

Research on language model pretraining focuses on architectural choices, optimization strategies, and data scaling laws [5, 6]. This work treats training data as given and optimizes model behavior. AI Visibility addresses the inverse problem: how should data be structured to optimize for learnability, independent of model architecture?

6.4 Prompt Engineering and Retrieval Augmentation

Recent work on prompt engineering [7] and retrieval-augmented generation [8] demonstrates methods for improving model outputs through better queries or external knowledge. These approaches operate post-training and assume learned representations already exist. AI Visibility addresses upstream conditions that determine what representations form during training.

7 Conclusion

This paper has introduced AI Visibility as a formal discipline addressing the upstream conditions that determine information learnability in large language model training. We have provided a canonical definition, established clear scope and boundaries, and developed a theoretical framework comprising seven interconnected theorems.

The framework demonstrates that failures commonly attributed to model limitations, prompt quality, or retrieval systems frequently originate from upstream signal degradation during content aggregation and compression. By formalizing principles of semantic stability, authorship determinism, and structural coherence, AI Visibility provides actionable guidance for information design that optimizes for machine learning rather than human ranking.

Key contributions include:

1. A formal definition distinguishing AI Visibility from search engine optimization, prompt engineering, and adjacent disciplines

2. Explicit non-equivalence conditions preventing semantic substitution by downstream tools or optimization products

3. A unified theoretical framework explaining how information becomes learnable through aggregation, stability, and provenance

4. Falsifiable predictions enabling empirical validation

7.1 Future Work

Immediate opportunities for extending this work include:

- **Empirical validation**: Systematic testing of framework predictions through controlled experiments on model training and recall

- **Operational patterns**: Development of concrete implementation patterns for AI Visibility principles across content types
- **Measurement frameworks**: Design of metrics for assessing semantic stability, authorship determinism, and aggregation coherence
- **Cross-model generalization**: Investigation of whether AI Visibility principles transfer across model families and architectures

As large language models continue to mediate information access, the need for formal frameworks addressing upstream information design will only increase. AI Visibility provides a foundation for systematic investigation of how content should be authored, structured, and maintained to ensure reliable machine learning and recall.

References

[1] Manning, C. D., Raghavan, P., & Schütze, H. (2008). *Introduction to Information Retrieval*. Cambridge University Press.

[2] Page, L., Brin, S., Motwani, R., & Winograd, T. (1999). *The PageRank Citation Ranking: Bringing Order to the Web*. Stanford InfoLab Technical Report.

[3] Berners-Lee, T., Hendler, J., & Lassila, O. (2001). *The Semantic Web*. Scientific American, 284(5), 34-43.

[4] Bollacker, K., Evans, C., Paritosh, P., Sturge, T., & Taylor, J. (2008). *Freebase: A Collaboratively Created Graph Database for Structuring Human Knowledge*. Proceedings of SIGMOD.

[5] Brown, T. B., et al. (2020). *Language Models are Few-Shot Learners*. Advances in Neural Information Processing Systems, 33.

[6] Kaplan, J., et al. (2020). *Scaling Laws for Neural Language Models*. arXiv preprint arXiv:2001.08361.

[7] Liu, P., et al. (2023). *Pre-train, Prompt, and Predict: A Systematic Survey of Prompting Methods in Natural Language Processing*. ACM Computing Surveys, 55(9).

[8] Lewis, P., et al. (2020). *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*. Advances in Neural Information Processing Systems, 33.